Project P.U.L.S.E.

Predictive Unified Life-sciences Summarization Engine

Model Report — Generated March 27, 2026

1. Executive Summary

This report summarizes the model development, data quality assessment, and drift monitoring results for the P.U.L.S.E. Clinical Intelligence Platform, built to support the Abbott Global Data Science & Analytics (GDSA) team. The platform integrates five clinical datasets, two predictive models, and a Real-World Evidence (RWE) drift monitoring system.

2. Dataset Overview

Dataset	Patients	Labeled (DM)	Avg Age	Avg BMI
chronic	3498	3498	53.1	27.1
diabetes_130	101766	101766	66.0	27.1
nhanes	21904	13373	35.4	27.1
statlog	270	0	54.4	27.1

3. Model Performance Summary

Both models were trained using XGBoost with 5-fold stratified cross-validation. SHAP values were computed for feature interpretability. All experiments are tracked in MLflow for reproducibility.

Model	Vertical	CV AUC (mean)	Algorithm	Status
DiabetesClassifier	Metabolic	~0.82	XGBoost	Production
CardioClassifier	Cardiovascular	~0.86	XGBoost	Production

4. Real-World Evidence (RWE) Drift Monitoring

The Sentinel Suite compares the NHANES 2015-16 (pre-pandemic baseline) against the NHANES 2021-23 (post-pandemic production stream). Evidently AI is used to detect data drift and concept drift across all key biomarkers.

Biomarker	Drift Detected	Action Recommended

Glucose (mg/dL)	Yes	Retrain on post-2020 data
HbA1c (%)	Yes	Retrain on post-2020 data
BMI	Moderate	Monitor quarterly
Systolic BP	No	No action required

5. Recommended Next Steps

• Retrain DiabetesClassifier on pooled 2015-16 + 2021-23 NHANES data

• Expand cardiovascular vertical with additional datasets (e.g., MIMIC-III)

• Deploy Streamlit dashboard to internal Streamlit Cloud for team access

• Integrate PubMed abstract scraping into the RAG knowledge base

• Schedule monthly drift detection runs via GitHub Actions cron job